

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: Theory of Statistics

(Session: 2025-26 Semester: II)

Problem Set 8

(Bayesian Estimation)

1. A report says that on an average, within a day, 50% of emails received are spam emails. A certain software claims that it can detect 99% of spam emails, and the probability for a false positive (non-spam detected as spam) is 5%. On a given day, what is the probability that a non-spam email is detected as spam?
2. Suppose you are trying to estimate how likely you are to catch a Pikachu when you throw a Poké Ball. Let θ be the probability of successfully catching Pikachu on a single attempt. From your past experience, you believe $\theta \sim \text{Beta}(2,2)$, the prior distribution. To validate your belief, you try catching Pikachu 10 times with 7 successes (caught) and 3 failures. What is the posterior distribution of $(\theta|data)$? What are the prior mean (old estimate of θ) and the posterior mean (the revised estimate)?
3. Let $X|\beta \sim \text{Gamma}(\alpha, 1/\beta)$, α is known and β (unknown) has the prior $\text{Gamma}(\lambda, \gamma)$. Is it a conjugate prior?
4. Let $X_1, \dots, X_n$ be a random sample from $\text{Neg Bin}(r, \theta)$, where $r(> 0)$ is a known integer. The sample space $\mathfrak{X} = \{0, 1, \dots\}$ and $\theta \in (0,1)$. The prior distribution is $\text{Beta}(\alpha, \beta)$, α, β are known. Is it a conjugate prior?
5. Let $X|\theta \sim \text{Bin}(n, \theta)$, $\theta \in (0,1)$. If $\pi(\theta) = 1/\theta(1 - \theta)$, $\theta \in (0,1)$ is an improper prior, prove or disprove that the posterior $\pi(\theta|x)$ is also improper
6. Let $X_1, \dots, X_n$ be a random sample from $P(\theta)$, $\theta > 0$. If $\pi(\theta) = 1/\theta$, $\theta > 0$ is an improper prior, prove or disprove that the posterior $\pi(\theta|x)$ is also improper.
7. Let X be a random observation from the population $U[0, \theta]$, $\theta > 0$. The prior density function is $\Pi(\theta) = \beta^2 \theta e^{-\beta \theta}$. Find the Bayes estimator of θ wrt the loss $L(\theta, d) = (\theta - d)^2$.
8. Let $X_1, \dots, X_n$ be a random sample from $\text{Exp}\left(\frac{1}{\theta}\right)$, $\theta > 0$. The prior distribution is $\text{Gamma}(\alpha, \beta)$, α, β are known. Find the Bayes estimators of (a) $g(\theta) = \theta$ and (b) $g(\theta) = e^{-\theta}$ under the loss function $L(\theta, d) = (g(\theta) - d)^2/\theta$.

9. The number of defects in a magnetic tape follows the Poisson (θ) distribution. The prior distribution of θ is a Gamma ($\alpha = 3, \beta = 1$), where α and β are the shape and rate parameters, respectively. Five rolls of magnetic tape are tested for defects and it is found that the number of defects is 2, 2, 6, 0 and 3. If we use the squared error loss function, what is the Bayes estimate of θ ?
10. Heights of individuals in a population have a Normal ($\theta, 2^2$) distribution. The prior distribution of θ is Normal (mean = 68", s.d. = 1"). Ten individuals are selected from the population at random and their average height is found to be 69.5". What is the Bayes estimate of θ , if the squared (absolute) error loss function is used?
11. Let X be a random observation from the population $N(\theta, 1)$ $\theta \in \mathfrak{R}$. The prior distribution is $N(0, \sigma^2)$, σ^2 known. Find the Bayes risk of the Bayes estimator under a squared error loss.
12. Let $X|\theta \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1)$ and θ follows the Jeffrey's prior. Find the Bayes estimator of $g(\theta) = \theta(1 - \theta)$ wrt the squared error loss function.
13. Let X be a random sample from $N(\theta, 1)$, $\theta \in \mathfrak{R}$. The improper prior is $\pi(\theta) = e^\theta$, $\theta \in \mathfrak{R}$. Find the generalized Bayes estimator under the squared error loss.
14. In a poll conducted among a sample of 2500 university students, 1125 responses were 'yes' and the remaining 'no'. A Bayesian is trying to estimate the proportion of 'yes', say, p , in the population assuming a beta prior. Calculate the 96% credible interval of p with the $\text{Beta}(1, 1)$ prior.
(Note: Though it's a non-normal and large sample case, obtain the exact credible interval and not the asymptotic, i.e., normal approximation one. Hint: You can use a well-known inequality in this case.)
15. $X_1, \dots, X_n$ be a random sample from $\text{Exp}\left(\frac{1}{\theta}\right)$, $\theta > 0$. If $\pi(\theta) \propto 1$, $\theta > 0$ is an improper prior, obtain an approximate $100(1 - \alpha)\%$ credible interval for θ .

Note: 1. Gamma distribution specified by Gamma (shape, scale), unless otherwise mentioned.
 2. Exponential distribution specified by Exponential (scale).
 3. Normal distribution is specified by Normal (mean, variance)